

Linear Inequalities — Full Solutions

Digital SAT Style | Every Step Fully Explained

Q1. Correct Answer: D) 4

STEP-BY-STEP SOLUTION:

$$4x - 3 > 9$$

$$4x > 12$$

$$x > 3$$

Only D=4 satisfies $x > 3$. Check: $4(4) - 3 = 13 > 9$ ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 1 → $4(1) - 3 = 1$. Not > 9 .

B) 2 → 5. Not > 9 .

C) 3 → 9. NOT > 9 (strict inequality excludes boundary).

D) 4 — CORRECT: $13 > 9$.

 **TRAP ALERT:** Picking C) $x=3$ because $4(3) - 3 = 9$. But $>$ is STRICT — 9 is NOT greater than 9.

 **KEY INSIGHT:** Strict inequality ($>$) excludes the boundary value. Always check $>$, not $\geq$.

 **FAST METHOD:** Solve $x > 3$. Only option > 3 is D.

Q2. Correct Answer: C) 6

STEP-BY-STEP SOLUTION:

$$48 + 4q \geq 70$$

$$4q \geq 22$$

$$q \geq 5.5$$

Since q must be a whole number, minimum $q = 6$.

Verify: $48 + 4(6) = 72 \geq 70$ ✓. But $48 + 4(5) = 68 < 70$ ✗.

WHY EACH OPTION IS WRONG/RIGHT:

A) 4 → $48 + 16 = 64 < 70$. Fails.

B) 5 → $48 + 20 = 68 < 70$. Fails.

C) 6 — CORRECT.

D) 7 → Works but not the minimum.

 **TRAP ALERT:** Getting $q \geq 5.5$ and rounding DOWN to 5. For "minimum number needed," always ROUND UP.

 **KEY INSIGHT:** When the solution is non-integer and you need a minimum count, round UP to the next whole number.

 **FAST METHOD:** $22/4 = 5.5 \rightarrow$ round up $= 6$.

Q3. Correct Answer: A) 3

STEP-BY-STEP SOLUTION:

$$2x + 5 \leq 13$$

$$2x \leq 8$$

$$x \leq 4$$

Check options: A) $3 \leq 4$ ✓. B) $5 \leq 4$ ✗. C) $6 \leq 4$ ✗. D) $7 \leq 4$ ✗.

WHY EACH OPTION IS WRONG/RIGHT:

A) 3 — CORRECT.

B) 5 → $5 \leq 4$ is false.

C) 6 → $6 \leq 4$ is false.

D) 7 → $7 \leq 4$ is false.

 **TRAP ALERT:** Solving the inequality correctly but not checking each option. Only A satisfies $x \leq 4$.

 **KEY INSIGHT:** Solve first, then test each option against the solution. Don't assume — verify.

 **FAST METHOD:** $x \leq 4$. Only $3 \leq 4$. Done.

Q4. Correct Answer: B) $x > -4$

STEP-BY-STEP SOLUTION:

$$-3x < 12$$

Divide by -3 . FLIP the inequality sign:

$$x > -4$$

WHY EACH OPTION IS WRONG/RIGHT:

A) $x < -4$ — Forgot to flip the sign when dividing by negative.

B) $x > -4$ — CORRECT.

C) $x < 4$ — Used positive 3 instead of -3 .

D) $x > 4$ — Neither flipped nor handled the negative.

 **TRAP ALERT:** THE #1 inequality error: forgetting to flip the sign when dividing by a negative number.

 **KEY INSIGHT:** Multiplying or dividing by a NEGATIVE number always FLIPS the inequality. Write "FLIP" in your margin.

 **FAST METHOD:** $-3x < 12 \rightarrow$ divide by $-3 \rightarrow$ FLIP $\rightarrow x > -4$.

Q5. Correct Answer: B)

STEP-BY-STEP SOLUTION:

$$\text{Total cost} = 80 + 0.25m$$

Condition: no more than \$200 $\rightarrow \leq$

$$80 + 0.25m \leq 200$$

WHY EACH OPTION IS WRONG/RIGHT:

A) Uses $<$ instead of $\leq$. "No more than" includes equality.

B) CORRECT.

C) Wrong direction (over budget region).

D) Wrong direction.

 **TRAP ALERT:** Confusing "no more than" ($\leq$) with "less than" ($<$). "No more than" includes exactly \$200.

 **KEY INSIGHT:** "At most" / "no more than" / "maximum" $\rightarrow \leq$. "At least" / "no less than" / "minimum" $\rightarrow \geq$.

 **FAST METHOD:** Keyword: "no more than" $\rightarrow \leq$. Done.

Q6. Correct Answer: Any integer from -2 to 5

STEP-BY-STEP SOLUTION:

$$-4 \leq 2x \leq 10$$

Divide all three parts by 2:

$$-2 \leq x \leq 5$$

Valid integers: $-2, -1, 0, 1, 2, 3, 4, 5$

WHY EACH OPTION IS WRONG/RIGHT:

Any value outside $[-2, 5]$ is wrong.

Common error: only solving one part of the compound inequality.

Any integer in range is CORRECT.

 **TRAP ALERT:** Solving only one inequality and ignoring the other bound.

 **KEY INSIGHT:** Compound inequalities: divide ALL parts simultaneously. If dividing by a negative, flip BOTH inequality signs.

 **FAST METHOD:** $-4/2 \leq x \leq 10/2 \rightarrow -2 \leq x \leq 5$. Enter 0.

Q7. Correct Answer: A)

STEP-BY-STEP SOLUTION:

"Less than \$12" $\rightarrow$ strict inequality $<$

$$\text{Cost} = 1.5b$$

$$1.5b < 12$$

WHY EACH OPTION IS WRONG/RIGHT:

A) CORRECT.

B) Uses $\leq$ but "less than" means strict $<$.

C) Wrong direction.

D) Wrong setup — divides instead of multiplies.

 **TRAP ALERT:** Using $\leq$ because the answer comes out clean. "Less than" always means $<$.

 **KEY INSIGHT:** "Spend less than X" $\rightarrow <$. "Spend at most X" $\rightarrow \leq$.

 **FAST METHOD:** "Less than" $\rightarrow <$. Cost = $1.5b$. Answer: $1.5b < 12$.

Q8. Correct Answer: A) 3

STEP-BY-STEP SOLUTION:

$$5x - 1 < 19$$

$$5x < 20$$

$$x < 4$$

Largest INTEGER strictly less than 4 is 3.

WHY EACH OPTION IS WRONG/RIGHT:

- A) 3 — CORRECT.
- B) 4 → $x=4: 5(4)-1=19$, not STRICTLY less than 19.
- C) 5 → $24 > 19$.
- D) 6 → $29 > 19$.

 **TRAP ALERT:** Choosing 4 because $x < 4$ "almost includes 4." Strict $<$ means 4 itself is EXCLUDED.

 **KEY INSIGHT:** $x < n$: largest integer = $n-1$. $x \leq n$: largest integer = n .

 **FAST METHOD:** $x < 4 \rightarrow$ largest integer = 3.

Q9. Correct Answer: C) 3

STEP-BY-STEP SOLUTION:

$$3(x + 4) \geq 21$$

$$x + 4 \geq 7$$

$$x \geq 3$$

Smallest integer ≥ 3 is 3.

WHY EACH OPTION IS WRONG/RIGHT:

- A) 1 → $1 \geq 3$ is false.
- B) 2 → $2 \geq 3$ is false.
- C) 3 — CORRECT: $3 \geq 3$ is true ($\geq$ includes boundary).
- D) 4 → Works but not the smallest.

 **TRAP ALERT:** Thinking $\geq$ excludes the boundary. $x \geq 3$ INCLUDES $x = 3$.

 **KEY INSIGHT:** $\geq$ includes the endpoint. The smallest integer satisfying $x \geq 3$ is exactly 3.

 **FAST METHOD:** $x \geq 3$. Smallest integer = 3.

Q10. Correct Answer: D) $x < 5$

STEP-BY-STEP SOLUTION:

Open dot = strict (not included)

Shading left = less than

Open dot at 5, shading left → $x < 5$

WHY EACH OPTION IS WRONG/RIGHT:

- A) $x \geq 5$ — Closed dot, shading right.
- B) $x > 5$ — Open dot, shading right (wrong direction).
- C) $x \leq 5$ — Closed dot, shading left.
- D) $x < 5$ — CORRECT.

 **TRAP ALERT:** Choosing C ($x \leq 5$) — the open dot means strict inequality, not $\leq$.

 **KEY INSIGHT:** Filled dot = $\leq$ or $\geq$. Open dot = $<$ or $>$. Left shading = "less than." Right = "greater than."

 **FAST METHOD:** Open+left = $x < 5$.

Q11. Correct Answer: A) $-8 < x \leq 12$

STEP-BY-STEP SOLUTION:

Inequality 1: $2x-7 < 3x+1 \rightarrow -8 < x \rightarrow x > -8$

Inequality 2: $x-4 \leq 8 \rightarrow x \leq 12$

Combined (AND): $-8 < x \leq 12$

WHY EACH OPTION IS WRONG/RIGHT:

A) $-8 < x \leq 12$ — CORRECT (strict at -8 , includes 12).

B) $x > -8$ — Missing the upper bound.

C) $x \leq 12$ — Missing the lower bound.

D) $-8 \leq x \leq 12$ — Wrong: -8 is strict ($<$), not $\leq$.

 **TRAP ALERT:** Writing $\leq$ at -8 instead of $<$. The first inequality gives $x > -8$ (strict).

 **KEY INSIGHT:** "AND" = intersection. Solve each separately, then combine. Watch strict vs non-strict at each boundary.

 **FAST METHOD:** $x > -8$ AND $x \leq 12 \rightarrow -8 < x \leq 12$.

Q12. Correct Answer: D) $y > (2/3)x - 4$

STEP-BY-STEP SOLUTION:

Region is ABOVE the line $\rightarrow y$ is greater $\rightarrow >$ or $\geq$

Boundary is DASHED $\rightarrow$ strict inequality $\rightarrow >$

Combined: $y > (2/3)x - 4$

WHY EACH OPTION IS WRONG/RIGHT:

A) $y \leq$ — Below, includes boundary. Wrong region.

B) $y <$ — Below, strict. Wrong region.

C) $y \geq$ — Above but solid boundary. Dashed means strict.

D) $y >$ — CORRECT: above + dashed.

 **TRAP ALERT:** Choosing C because the region is above — forgetting that dashed = strict ($>$ not $\geq$).

 **KEY INSIGHT:** Two checks: (1) above or below? (2) solid or dashed? Above+dashed = $y >$.

 **FAST METHOD:** Above + dashed $\rightarrow y >$ line.

Q13. Correct Answer: D) 11

STEP-BY-STEP SOLUTION:

Maximize $p - q$: use $\max(p)$ and $\min(q)$.

$\max(p) = 8$, $\min(q) = -3$

$p - q = 8 - (-3) = 8 + 3 = 11$

WHY EACH OPTION IS WRONG/RIGHT:

A) 5 — Used $\max(p) - \max(q) = 8 - 3 = 5$.

B) 8 — Just $\max(p)$ alone.

C) 10 — Used $\max(p) - \min(|q|) = 8 - (-2) = 10$.

D) 11 — CORRECT.

 **TRAP ALERT:** Using $\max$ of both: $8 - 3 = 5$. To maximize a DIFFERENCE, minimize the subtracted term.

💡 **KEY INSIGHT:** $\text{Max}(a-b) = \text{max}(a) - \text{min}(b)$. $\text{Min}(a-b) = \text{min}(a) - \text{max}(b)$.

⚡ **FAST METHOD:** $8 - (-3) = 11$.

Q14. Correct Answer: B) 3

STEP-BY-STEP SOLUTION:

Remaining calories: $2400 - 520 - 680 = 1200$

Each serving: 320 cal

$1200/320 = 3.75$

Maximum FULL servings: 3 (cannot eat 0.75 of a serving)

Verify: $3 \times 320 = 960 \leq 1200$ ✓. $4 \times 320 = 1280 > 1200$ ✗.

WHY EACH OPTION IS WRONG/RIGHT:

A) 2 → $640 < 1200$, room for more.

B) 3 — CORRECT.

C) 4 → $1280 > 1200$. Over limit.

D) 5 → Way over.

👾 **TRAP ALERT:** Dividing 2400 by 320 directly ($=7.5$) without subtracting breakfast and lunch first.

💡 **KEY INSIGHT:** Subtract consumed calories first, then divide remaining by per-serving. Floor the result.

⚡ **FAST METHOD:** $1200/320 = 3.75 \rightarrow \text{floor} = 3$.

Q15. Correct Answer: A) $-1 \leq x \leq 7$

STEP-BY-STEP SOLUTION:

$|2x-6| \leq 8$ means $-8 \leq 2x-6 \leq 8$

Add 6: $-2 \leq 2x \leq 14$

Divide by 2: $-1 \leq x \leq 7$

WHY EACH OPTION IS WRONG/RIGHT:

A) $-1 \leq x \leq 7$ — CORRECT.

B) $x \leq -1$ or $x \geq 7$ — This is $|2x-6| \geq 8$ (wrong direction).

C) $-7 \leq x \leq 1$ — Subtracted 6 from wrong side.

D) $x \leq 7$ — Only the right bound, missing left.

👾 **TRAP ALERT:** B is the solution to $|2x-6| \geq 8$. Classic sign confusion with absolute value inequalities.

💡 **KEY INSIGHT:** $|\text{expr}| \leq k \rightarrow \text{AND (between)}$. $|\text{expr}| \geq k \rightarrow \text{OR (outside)}$.

⚡ **FAST METHOD:** $|2x-6| \leq 8 \rightarrow -8 \leq 2x-6 \leq 8 \rightarrow \text{add 6} \rightarrow \text{divide 2}$.

Q16. Correct Answer: 1

STEP-BY-STEP SOLUTION:

Through $(0, -2)$ and $(3, 0)$. Slope $= (0 - (-2))/(3 - 0) = 2/3$.

Line: $y = (2/3)x - 2$. Standard form: $2x - 3y = 6$.

Rewrite to match $rx + ty \geq -6$: multiply by -1 : $-2x + 3y = -6$.

So: $-2x+3y \geq -6 \rightarrow r=-2, t=3$.

$r+t = -2+3 = 1$.

Verify with (0,0): $-2(0)+3(0)=0 \geq -6$ ✓ (origin is in the solution region, above the line).

WHY EACH OPTION IS WRONG/RIGHT:

$r+t=5 \rightarrow$ added $2+3$ without checking signs.

$r+t=-5 \rightarrow$ flipped all signs wrong.

$r+t=-1 \rightarrow$ sign error on t .

1 — CORRECT.

 **TRAP ALERT:** Finding $2x-3y=6$ and using $r=2, t=-3$. Must rearrange to match the form with -6 on the right.

 **KEY INSIGHT:** Match the EXACT form given in the problem. Rearrange systematically.

 **FAST METHOD:** Line: $2x-3y=6$. Multiply by -1 : $-2x+3y=-6$. $r=-2, t=3$. Sum=1.

Q17. Correct Answer: C) 10

STEP-BY-STEP SOLUTION:

$$250 + 65h \leq 900$$

$$65h \leq 650$$

$$h \leq 10$$

Maximum full hours = 10.

Verify: $250+65(10)=250+650=\$900 \leq \900 ✓. 11 hrs: $\$965 > \900 ✗.

WHY EACH OPTION IS WRONG/RIGHT:

A) 8 $\rightarrow$ \$770. Under budget but not maximum.

B) 9 $\rightarrow$ \$835. Under budget but not maximum.

C) 10 — CORRECT: exactly at budget.

D) 11 $\rightarrow$ \$965. Over budget.

 **TRAP ALERT:** Thinking \$900 exactly is not allowed. "At most" includes equality: $\$900 \leq \900 is TRUE.

 **KEY INSIGHT:** "At most \$900" = $\leq \$900$. The exact boundary IS included.

 **FAST METHOD:** $65h \leq 650 \rightarrow h \leq 10$. Max full hours=10.

Q18. Correct Answer: D) -3

STEP-BY-STEP SOLUTION:

$$3y-2 > 4y+1 \rightarrow -y > 3 \rightarrow y < -3 \text{ (flip when dividing by } -1)$$

Solutions: all y strictly less than -3 .

Check each: A) $-4 < -3$ ✓. B) $-5 < -3$ ✓. C) $-6 < -3$ ✓. D) $-3 < -3$? NO.

$y=-3$ is NOT a solution (boundary excluded by strict $<$).

WHY EACH OPTION IS WRONG/RIGHT:

A) -4 — IS a solution. Not the answer.

B) -5 — IS a solution.

C) -6 — IS a solution.

D) -3 — NOT a solution. CORRECT answer to "which is NOT."

 **TRAP ALERT:** Solving correctly but choosing a value that IS a solution instead of the one that isn't.

 **KEY INSIGHT:** "Which is NOT" questions: solve the inequality, then find which option fails it. Boundaries of strict inequalities always fail.

 **FAST METHOD:** $y < -3$. Boundary $y = -3$ fails. Answer D.

Q19. Correct Answer: D) (0, 6)

STEP-BY-STEP SOLUTION:

$y > -(3/4)x + 5$. Substitute each:

A) (4,2): $\text{RHS} = -3 + 5 = 2$. $2 > 2$? NO (strict). Fails.

B) (0,3): $\text{RHS} = 5$. $3 > 5$? NO. Fails.

C) (0,4): $\text{RHS} = 5$. $4 > 5$? NO. Fails.

D) (0,6): $\text{RHS} = 5$. $6 > 5$? YES ✓.

WHY EACH OPTION IS WRONG/RIGHT:

A) Boundary point — excluded by strict $>$.

B) $3 > 5$ is false.

C) $4 > 5$ is false.

D) $6 > 5$ is true. CORRECT.

 **TRAP ALERT:** Choosing A because $2 = 2$, forgetting strict $>$ excludes the boundary.

 **KEY INSIGHT:** Test $x = 0$ options first — they require only checking the y -intercept comparison. Fastest to verify.

 **FAST METHOD:** At $x = 0$: $\text{RHS} = 5$. Need $y > 5$. Only D has $y = 6 > 5$.

Q20. Correct Answer: 4

STEP-BY-STEP SOLUTION:

Condition 1: $2n - 1 < 15 \rightarrow n < 8 \rightarrow$ positive integers: $\{1, 2, 3, 4, 5, 6, 7\}$

Condition 2: $n^2 > 9 \rightarrow n > 3 \rightarrow$ positive integers: $\{4, 5, 6, \dots\}$

Intersection: $\{4, 5, 6, 7\}$

Count = 4

WHY EACH OPTION IS WRONG/RIGHT:

3 — Forgot $n = 7$ satisfies both.

7 — Only applied condition 1.

5 — Off-by-one at boundary.

4 — CORRECT.

 **TRAP ALERT:** Solving $n > 3$ as $n \geq 4$ correctly but then miscounting the intersection.

 **KEY INSIGHT:** List each solution set explicitly, then find the overlap. Don't skip this step.

 **FAST METHOD:** $n \in \{1..7\}$ AND $n \in \{4, 5, \dots\} \rightarrow \{4, 5, 6, 7\} = 4$ values.

Q21. Correct Answer: C) $a \geq 16$

STEP-BY-STEP SOLUTION:

"At least 16 years old" means $\text{age} \geq 16$.

This allows entry WITHOUT a consent form.

WHY EACH OPTION IS WRONG/RIGHT:

- A) $a < 16$ — These ages NEED a consent form.
- B) $a \leq 16$ — Includes ages below 16.
- C) $a \geq 16$ — CORRECT.
- D) $a > 16$ — Excludes exactly 16, but 16 IS old enough.

 **TRAP ALERT:** Choosing D because "at least 16" sounds like "more than 16." It includes 16.

 **KEY INSIGHT:** "At least N" = $\geq N$ (includes N). "More than N" = $> N$ (excludes N).

 **FAST METHOD:** "At least 16" $\rightarrow a \geq 16$.

Q22. Correct Answer: A) -30

STEP-BY-STEP SOLUTION:

Check all four corner products:

$$(-5)(-2) = 10$$

$$(-5)(6) = -30$$

$$(4)(-2) = -8$$

$$(4)(6) = 24$$

$$\text{Minimum} = -30$$

WHY EACH OPTION IS WRONG/RIGHT:

- A) -30 — CORRECT: $m=-5, n=6$.
- B) -20 — Not a corner product.
- C) -10 — Not a corner product.
- D) -8 — $m=4, n=-2$. Not the minimum.

 **TRAP ALERT:** Only checking $m=4, n=-2$ ($=-8$) and missing the more negative $m=-5, n=6$ ($=-30$).

 **KEY INSIGHT:** Min/max of a product: always check ALL FOUR corner combinations.

 **FAST METHOD:** $(-5)(6)=-30$ is the most negative.

Q23. Correct Answer: C) Quadrants III and IV

STEP-BY-STEP SOLUTION:

Intersection: $x+1=-2x+4 \rightarrow 3x=3 \rightarrow x=1, y=2$ (in QI).

Test QI (1,2): $2 \geq 2$ ✓, $2 \leq 2$ ✓. Solutions exist in QI.

Test QII (-1,3): $3 \geq 0$ ✓, $3 \leq 6$ ✓. Solutions exist in QII.

Test QIII (-2,-2): $-2 \geq -1$? NO. No solutions in QIII.

Test QIV (2,-1): $-1 \geq 3$? NO. No solutions in QIV.

Quadrants III and IV have NO solutions.

WHY EACH OPTION IS WRONG/RIGHT:

- A) QII only — QII has solutions (tested above).
- B) QIV only — QIII also has no solutions.
- C) III and IV — CORRECT.
- D) QIII only — QIV also has no solutions.

 **TRAP ALERT:** Not testing enough points. Always test one point per quadrant.

 **KEY INSIGHT:** For system-of-inequalities quadrant questions: test one representative point per quadrant. Use Desmos on test day!

 **FAST METHOD:**  DESMOS: Graph $y \geq x+1$ and $y \leq -2x+4$. Visually confirm which quadrants have shading.

Q24. Correct Answer: B) 2

STEP-BY-STEP SOLUTION:

$$kx+4 > 2x+k \rightarrow (k-2)x > k-4$$

For ALL real x to satisfy this: coefficient of x must be 0 AND constant must be true.

$$k-2=0 \rightarrow k=2$$

Check constant: $0 > 2-4 = -2$. $0 > -2$ is TRUE for all x .

So $k=2$ makes the inequality universally true.

WHY EACH OPTION IS WRONG/RIGHT:

A) $0 \rightarrow -2x > -4 \rightarrow x < 2$. Not all x .

B) 2 — CORRECT.

C) $4 \rightarrow 2x > 0 \rightarrow x > 0$. Not all x .

D) $6 \rightarrow 4x > 2 \rightarrow x > 0.5$. Not all x .

 **TRAP ALERT:** Thinking "solutions for all x " means "any k works." Only $k=2$ eliminates the x -term entirely.

 **KEY INSIGHT:** For an inequality to hold for ALL x : the coefficient of x must vanish, and the resulting constant inequality must be true.

 **FAST METHOD:** Set $k-2=0 \rightarrow k=2$. Check: $0 > -2$. TRUE. Done.

Q25. Correct Answer: A)

STEP-BY-STEP SOLUTION:

Machine time: 3 hrs/unit X + 5 hrs/unit $Y \leq 60$

Minimum Y production: $y \geq 4$

Non-negativity: $x \geq 0$

WHY EACH OPTION IS WRONG/RIGHT:

A) CORRECT.

B) ≥ 60 would mean at LEAST 60 hours. Wrong direction.

C) Swaps 3 and 5 (hours per product).

D) $y \leq 4$ means at MOST 4, not at LEAST 4.

 **TRAP ALERT:** Swapping which product takes 3 vs 5 hours, or reversing $y \geq 4$ to $y \leq 4$.

 **KEY INSIGHT:** Read each constraint separately: (1) resource limit $\rightarrow \leq$, (2) minimum production $\rightarrow \geq$, (3) non-negativity $\rightarrow \geq 0$.

 **FAST METHOD:** Match keywords: "at most 60 hours" $\rightarrow \leq 60$, "at least 4 units" $\rightarrow \geq 4$.

Q26. Correct Answer: A) $y \leq -2x + 3$

STEP-BY-STEP SOLUTION:

$$\text{Slope: } (-5-7)/(4-(-2)) = -12/6 = -2$$

Using point $(-2, 7)$: $y - 7 = -2(x + 2) \rightarrow y = -2x + 3$

Verify: $y = -2(-2) + 3 = 7$ ✓ and $y = -2(4) + 3 = -5$ ✓

Below + solid boundary $\rightarrow y \leq$

Answer: $y \leq -2x + 3$

WHY EACH OPTION IS WRONG/RIGHT:

A) CORRECT.

B) Positive slope — wrong (slope is -2).

C) y-intercept of -3 — calculation error.

D) Dashed boundary — problem says solid.

 **TRAP ALERT:** Computing the y-intercept wrong: $b = 7 - (-2)(-2) = 7 - 4 = 3$, NOT $7 + 4 = 11$.

 **KEY INSIGHT:** Point-slope: $y - y_1 = m(x - x_1)$. Expand carefully and verify with both given points.

 **FAST METHOD:** Slope $= -2$. y-int: plug $(0, y)$: $y = -2(0) + b \rightarrow$ use $(-2, 7)$: $7 = 4 + b \rightarrow b = 3$.
Below + solid $\rightarrow y \leq -2x + 3$.

Q27. Correct Answer: A)

STEP-BY-STEP SOLUTION:

$$4 - kx \geq 2 - 3x \rightarrow 2 \geq (k - 3)x$$

Case 1: $k > 3 \rightarrow (k - 3) > 0 \rightarrow$ divide normally $\rightarrow x \leq 2/(k - 3)$

Case 2: $0 < k < 3 \rightarrow (k - 3) < 0 \rightarrow$ divide and FLIP $\rightarrow x \geq 2/(k - 3)$

This matches option A.

WHY EACH OPTION IS WRONG/RIGHT:

A) CORRECT — accounts for both cases.

B) Ignores sign flip when $k < 3$.

C) Ignores that $k > 3$ gives $\leq$ not $\geq$.

D) Oversimplification.

 **TRAP ALERT:** Dividing by $(k - 3)$ without checking its sign. The direction of the inequality depends on whether $k > 3$ or $k < 3$.

 **KEY INSIGHT:** Dividing by an expression with an unknown: ALWAYS check both positive and negative cases.

 **FAST METHOD:** $2 \geq (k - 3)x$. Two cases based on sign of $(k - 3)$.

Q28. Correct Answer: 3,401

STEP-BY-STEP SOLUTION:

Need total votes $> 60\%$ of $10,000 = 6,000$.

Already has 2,600. Remaining available: 6,000.

$$2,600 + r > 6,000$$

$$r > 3,400$$

Minimum whole number: 3,401.

Verify: $2,600 + 3,401 = 6,001 > 6,000$ ✓. $2,600 + 3,400 = 6,000$ which is NOT more than 60%.

WHY EACH OPTION IS WRONG/RIGHT:

3,400 $\rightarrow$ gives exactly $6,000 = 60\%$, not MORE than 60%.

3,600 → works but not the minimum.

3,000 → only 56%.

3,401 — CORRECT.

 **TRAP ALERT:** "More than 60%" is STRICT. 60.00% is not enough. Minimum integer above 3,400 is 3,401.

 **KEY INSIGHT:** Strict $>$ with whole numbers: minimum integer satisfying $r > N$ is $N+1$.

 **FAST METHOD:** $r > 3400 \rightarrow \min = 3401$.

Q29. Correct Answer: A) $x \geq 1$

STEP-BY-STEP SOLUTION:

$$-3(2x-1) \leq 6x-9$$

$$\text{Distribute: } -6x + 3 \leq 6x - 9$$

$$\text{Add } 6x: 3 \leq 12x - 9$$

$$\text{Add } 9: 12 \leq 12x$$

$$\text{Divide: } 1 \leq x \rightarrow x \geq 1$$

WHY EACH OPTION IS WRONG/RIGHT:

A) $x \geq 1$ — CORRECT.

B) $x \leq 1$ — Wrong direction.

C) $x \geq 2$ — Arithmetic error: $12/12=1$, not 2.

D) $x \leq -1$ — Multiple errors.

 **TRAP ALERT:** Sign error distributing -3 : $-3(2x-1) = -6x+3$, NOT $-6x-3$.

 **KEY INSIGHT:** Distribute a negative: $-3(2x-1) = (-3)(2x) + (-3)(-1) = -6x+3$. The -1 becomes $+3$.

 **FAST METHOD:** $-6x+3 \leq 6x-9 \rightarrow 12 \leq 12x \rightarrow x \geq 1$.

Q30. Correct Answer: A) $a < 2$

STEP-BY-STEP SOLUTION:

For (a,b) to satisfy both:

$$b \geq 3a-1 \text{ (from inequality 1)}$$

$$b < -a+7 \text{ (from inequality 2)}$$

$$\text{For these to be compatible: } 3a-1 < -a+7$$

$$4a < 8 \rightarrow a < 2$$

This MUST be true for any solution (a,b) to exist.

WHY EACH OPTION IS WRONG/RIGHT:

A) $a < 2$ — CORRECT: necessary condition.

B) $b > 5$ — Not necessarily (e.g., $a=0$, $b=0$ satisfies both).

C) $a > 2$ — Opposite of truth.

D) $b < 5$ — Not necessarily ($b=6$ works when a is small).

 **TRAP ALERT:** Choosing B or D based on checking one specific point instead of finding what MUST be true for ALL solutions.

 **KEY INSIGHT:** "Must be true" = find the condition that ALL solution points satisfy. Set lower bound $<$ upper bound.

 **FAST METHOD:** $3a-1 < -a+7 \rightarrow 4a < 8 \rightarrow a < 2$.

TOPIC SUMMARY — Linear Inequalities

Common Mistakes:

1. Forgetting to FLIP the inequality when multiplying/dividing by a negative number.
2. Confusing "at most" ($\leq$) with "less than" ($<$), and "at least" ($\geq$) with "greater than" ($>$).
3. On graphs: using solid for dashed (or vice versa), causing $\geq$ vs $>$ errors.

Must Remember:

1. Keywords: "at least"/"minimum"/"no less than" $\rightarrow \geq$. "At most"/"maximum"/"no more than" $\rightarrow \leq$.
2. $|\text{expr}| \leq k \rightarrow -k \leq \text{expr} \leq k$ (AND/between). $|\text{expr}| \geq k \rightarrow \text{expr} \leq -k$ OR $\text{expr} \geq k$ (OR/outside).
3. Graph: above the line = $y >$ or $y \geq$. Below = $y <$ or $y \leq$. Dashed = strict. Solid = includes boundary.

Expected on DSAT: 2–4 questions per test. Expect one word-problem inequality, one graphical interpretation, and one compound/absolute value.

 **DESMOS:** For two-variable inequalities, type them directly (e.g., $y > 2x-3$). Desmos shades the solution region automatically. Test points visually — massive time saver on test day.